Approximation algorithms exercises

35-E01 Matching lower bounds and a weighted counterexample

A four-leaf star has center weight 1 and each leaf weight 100. Ignore weights first and give the size of the cover obtained by selecting both endpoints of any edge, together with OPT. Then compute weighted cost and weighted optimum for that same output. Prove why the cardinality matching bound gives no corresponding weighted approximation guarantee.

35-E02 Metric walks and tours

For A(0,0),B(3,0),C(3,2),D(0,2), use Manhattan costs and MST chain A-D-C-B. Give the complete double-tree walk, first-visit tour, and their costs. Independently prove the MST and TSP optima. Which proved lower bounds contradict a claimed TSP optimum of 6?

35-E03 Weighted set-cover charges

Let X={1,...,6}, S1={1,2,3}, S2={4,5,6}, S3={1,2,4,5}, with costs 2,2,1. Break equal-cost, equal-new-count ties in favor of S1. List cost per new element, choice, remaining elements, and each element's charge per round. Find OPT and verify the harmonic bound through charges on every optimum set.

35-E04 Expected objective versus expected ratio

Use clauses (a∨b∨c), (a∨¬b∨c), (¬a∨b∨c), (¬a∨¬b∨c). Under independent fair variables, find Y's distribution, E[Y], OPT, and E[OPT/Y]. Apply conditional expectation in a,b,c order, breaking ties toward 0. Give both branch values at each step and the final assignment. Why is independence of the four clause events unnecessary?

35-E05 LP bound, integer optimum, and rounding

A triangle has weights 3,4,5 on a,b,c. Prove the half-value LP solution optimal by finding nonnegative multipliers on its three edge constraints whose sum has left side $3x_a + 4x_b + 5x_c$. Find integer OPT, threshold-rounding cost, and actual approximation ratio.

Approximation algorithms exercises (continued)

35-E06 Four-round FPTAS integer trace

Use a=[102,98,195,101], t=305, ε=2/5 and δ=ε/(2n). List each merged unique list, trimmed list, and capacity-filtered list. Return a concrete subset. Find OPT by full subset enumeration or an exact list, and check feasibility and approximation. Do not use ε itself as the per-round tolerance.

35-E07 Representative direction and negative items

Why does the FPTAS invariant require smaller representatives? For a=[4,5,6], t=10, suppose after the first two items sum 4 is discarded and represented only by the larger sum 5. Trace the last item and identify the lost feasible sum. Use a=[8,-3], t=5 to disprove unrestricted early removal of sums above t for negative inputs. Distinguish failure of this proof from impossibility of every modified approximation algorithm.

35-E08 Gap reductions and adding constants

Assume deterministic factor-2 approximation for general TSP. For an n=5 HC input, assign original edges 1 and nonedges 11. Give yes/no cost ranges and the decision threshold. For any nonnegative complete graph, add K=max edge cost to every edge. Prove the new costs satisfy triangle inequality and preserve optimal tours, yet a ratio-ρ solution for the new costs need not give ratio ρ for the original costs.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

Original educational synthesis by Lin Cing. Original material is licensed under CC BY 4.0. The referenced textbook is not distributed with these notes.

https://github.com/xinhi6716/computer-science-self-study